

CLUB club

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Primary nucleotide archives

■ INSDC

the International Nucleotide Sequence Database Collaboration is comprised of

- **DDBJ**: DNA Data Bank of Japan.
- **EMBL-EBI**: European Bioinformatics Institute, part of the European Molecular Biology Laboratory (outside Cambridge, UK).
- **NCBI**: National Center for Biotechnology Information (US).

The INSDC has three levels of data:

- Raw reads. NCBI's SRA, the ENA read archive, and DDBJ's DRA.
- Assembled and annotated sequences. The "GenBank" records proper.
- Metadata. BioProject and BioSample, which link samples, projects, and runs.

These data are mirrored in all three centres. They also share a data policy: anyone can use the data for anything, without restriction.

Outside INSDC

Don't assume a search of NCBI is comprehensive.

GSA: China's CNCB-NGDC Genome Sequence Archive (GSA) has become a major repository. Its data don't automatically appear in GenBank or ENA.

GISAID: started for influenza in 2008 and became the dominant SARS-CoV-2 repository, with ~17.6 million sequences.

Pathoplexus: A fully-open alternative to GISAID launched in 2024. It syncs to INSDC, and NCBI Virus and is a virus-specific view over GenBank and RefSeq.

BV-BRC: Merged PATRIC, IRD, and ViPR. It adds consistent re-annotation and analysis tooling on top of INSDC bacterial and viral data.

There are also many curated organism-specific or community resources, such as HBVdb, LANL's HIV database, and Nextstrain's curated datasets.

INSDC curated and derived resources

The following datasets are derived from core INSDC assets. They are not mirrored.

■ RefSeq (NCBI)

Is a non-redundant, curated set of reference genomes, transcripts, and proteins derived from INSDC data.

■ Ensembl and Ensembl Genomes (EMBL-EBI)

Don't archive sequence. They take assemblies and run consistent annotation pipelines on them: gene models, comparative genomics, variation, and regulatory features. Ensembl covers vertebrates, and Ensembl Genomes covers plants, fungi, protists, metazoa, and bacteria.

■ UniProt (various)

Protein sequences and from three sources. **Swiss-Prot**, manually curated and small, is maintained by the SIB Swiss Institute of Bioinformatics (Geneva). **TrEMBL** mostly translated from INSDC coding sequences, is automatically annotated and enormous (EBI). **PIR**, the Protein Information Resource (Georgetown University Medical Center, Washington DC)

RefSeq

RefSeq accession IDs start with two letters and then have an underscore. The (unofficial) prefix meanings are

Prefix	Molecule
NC_	Complete genomic molecule: a chromosome, organelle genome, plasmid, or complete viral genome
NG_	Genomic region, e.g. a gene with flanking sequence, used for RefSeqGene and variant reporting
NT_, NW_	Genomic contigs and scaffolds; NW_ is for those derived from WGS assemblies
NZ_	Genomic records from unfinished WGS assemblies, chiefly prokaryotes
NM_	mRNA
NR_	Non-coding RNA
NP_	Protein

The first letter roughly separates curated or evidence-supported records from computed ones:

N: These carry the main reference records. **N** can't be read simply as "curated", though: many **NM**/**NP** records are generated by automated pipelines. The record's status field tells you more than the letter. **X**: The parallel **XM**, **XR**, and **XP** records are model predictions, mostly from NCBI's Gnomon gene-prediction pipeline. They are replaced by **N** records if supporting evidence is reviewed.

Other letters. A few prefixes break the two-letter logic:

- **AC** and **AP** are complete genomic molecules and proteins from alternate assemblies.
- **YP** is a protein record without a corresponding transcript record, typical of viral and organelle genomes.
- **WP** is a non-redundant protein: one record per identical protein sequence across many prokaryotic genomes, rather than one per genome.

NCBI Virus

NCBI Virus is a portal over GenBank and RefSeq. Sequences keep their GenBank or RefSeq accessions.

Consists of a specialised database holding sequence information from the annotation pipeline alongside sample metadata in standardised formats, plus a search interface and display tools. The valuable part for most users is metadata normalisation.

Its **VADR** (Viral Annotation DefineR) classifies submitted sequences against curated RefSeq models, then maps features from the matched RefSeq via a covariance-model alignment of the full sequence. It's built into GenBank's submission pipeline, so clean submissions are accepted and annotated with no indexer involvement, and it's open source for local use. Early versions excluded circular and segmented genomes - still true?

VADR's annotation becomes part of the GenBank record and propagates through INSDC, but the normalised metadata, the taxonomy grouping and the interface stay at NCBI. If you pull the same accessions from ENA you get the sequence and submitter annotation but not the tidied metadata.

Download is available in bulk with filters applied, and there's API and command-line access via NCBI Datasets. The SARS-CoV-2 Data Hub extends the model to raw SRA runs. For virus work the main caveats are that completeness is limited by what reaches INSDC – the GISAID and GSA gaps apply – and that normalised metadata is only as good as the submitter's original, so absent collection dates stay absent.

Using NCBI Virus command-line

Last night I discovered the NCBI datasets and dataformat (<https://www.ncbi.nlm.nih.gov/datasets/docs/v2/command-line-tools/download-and-install/>) tools.

Do this to install them (and `jq`):

```
$ pixi add ncbi-datasets-cli jq
```

Start reading here: https://www.ncbi.nlm.nih.gov/datasets/docs/v2/getting_started/